Hydrogen-combustion turbofans for practical zero-carbon aviation

Engines first → aircraft next | Compressed gaseous H_2 | Certification-first in FAA / Denver, CO. USA.

Investor Deck

The problem

Aviation needs deep decarbonization without breaking range, safety, or infrastructure.

- Jet fuel emissions face tightening regulation and rising cost exposure.
- Battery-electric flight is constrained by energy density, payload, and range.
- Many hydrogen concepts depend on cryogenic liquid H_2-complex, costly, and hard to certify.
- Airlines need a credible path they can adopt in steps: evidence → certification → deployment.

Our solution

Hydrogen combustion using compressed gaseous H_2: simpler operations, clearer certification path.

- Hydrogen-combustion turbofan architecture engineered for safety, repeatability, and certification-grade evidence.
- Compressed gaseous hydrogen avoids cryogenic cooling and reduces infrastructure burden in early deployments.
- Safety-first operations: leak detection, purge/vent routines, E-stop architecture, and auditable documentation.
- Built as a practical transition: engines first → aircraft next (only after propulsion proof is mature).

Two-phase roadmap

De-risk propulsion first, then scale to purpose-built aircraft.

Phase 1 - Engines

- Build hydrogen combustion subsystem → integrated prototype runs
- Endurance + repeatability milestones
- Certification evidence packages and safety case artifacts
- Early revenue: paid evaluations, prototype contracts, engineering + certification support

Phase 2 - Aircraft

- Purpose-built aircraft optimized around hydrogen combustion
- Integrated fuel storage + operations playbooks for airports
- Ground test campaigns → controlled demonstrations as authorized
- Scale production via partnerships and quality systems

Why Denver, Colorado (USA)

Primary headquarters for leadership, engineering, and regulatory strategy.

- Core executive leadership, engineering architecture, and systems integration based in Denver.
- Direct alignment with FAA regulatory pathways, certification strategy, and U.S. aerospace standards.
- Access to a deep aerospace and advanced engineering talent pool across Colorado and the U.S.
- Proximity to strategic partners, suppliers, and U.S.-based investors.
- Stable legal, corporate, and capital markets environment for long-term aviation programs.

Denver anchors governance, certification discipline, and commercialization.

Why Guanacaste, Costa Rica

Operations near Daniel Oduber Quirós Int'l Airport (LIR): practical testing, credibility, and access.

- Real aviation environment for safety discipline, logistics, and partner demonstrations (as authorized).
- Year-round operating conditions that reduce test schedule risk.
- Strong alignment with renewable energy narrative for hydrogen programs.
- Rented offices + leased workshop/warehouse: scalable footprint as program maturity grows.

Target market

Early focus on customers who need a certifiable decarbonization pathway.

- Regional airlines & cargo operators pursuing fleet decarbonization and cost stability.
- Engine & aircraft OEMs evaluating hydrogen propulsion options and retrofit pathways.
- Government / strategic aviation programs focused on energy security and emissions targets.
- Leasing companies and airport ecosystem partners preparing hydrogen readiness.

Competitive landscape

We win with a turbine-native path and reduced cryogenic dependence.

ZeroAvia

Combustion turbofan path; avoids fuel-cell complexity.

H2FLY

Targets turbofan-class propulsion & airline-scale adoption.

Airbus ZEROe

Faster phased execution; engines first, aircraft next.

SAF (biofuels)

Hydrogen fuel path targets true zero-carbon operation.

Battery-electric

Better range/payload vs battery constraints.

Cryogenic LH_2

Avoids cryogenic cooling and early infrastructure burden.

Business model

Multiple revenue lines in Phase 1, scaling to engines and aircraft programs.

- Engine programs: development milestones → deliveries as maturity increases.
- Prototype development contracts & engineering design services.
- Certification support services and safety documentation packages.
- Maintenance/service packages as deployments begin.
- Research partnerships + non-dilutive grants for clean aviation.

Go-to-market

Credibility-first: evidence, demonstrations, and partner programs.

Sales channels

- Direct enterprise sales to airlines, cargo ops, OEMs, and government programs.
- Partner-led deals via JDAs with aerospace suppliers and hydrogen ecosystem firms.
- Milestone contracts with staged payments (evaluation → prototype → program).

Marketing activities

- Invite-only demos at LIR facility + technical proof packages.
- Aviation conferences, targeted outreach, and executive briefings.
- White papers, safety documentation highlights, and partner co-marketing.

Financial outlook

Model-based growth from services → programs → engine sales as evidence matures.

$3.8M
Projected revenue (2027)

$79.9M
Projected revenue (2030)

$263.5M
Projected revenue (2031)

2030
Break-even year (model target)

$49.6M
Projected profit (2030)

$187.2M
Projected profit (2031)

Revenue mix evolves from paid evaluations + prototype contracts to scalable engine programs and deliveries.

Note: Figures reflect the current ColibriV financial model assumptions (2027–2031).

Funding plan

Milestone-based rounds to maintain runway and reduce risk.

Total planned equity: $47M

- Round 1 (Jan 2027): $12M - team + safe hydrogen test ops near LIR
- Round 2 (Jul 2028): $25M - scale testing + certification evidence + customer traction
- Round 3 (Mar 2029): $10M - delivery capacity + Phase 2 foundations

Use of funds

- Test infrastructure & safety systems (evidence factory)
- Propulsion + hydrogen engineering team build-out
- Prototype builds, endurance testing, supplier qualification
- Certification artifacts, traceability, and regulatory engagement
- Customer evaluation programs and partner demonstrations

Milestones

Monthly execution in 2027–2028; annual gates after program maturity.

2027–2028 (selected monthly)

- Jan 2027: Round 1 closes; safety program kickoff
- Mar 2027: Hydrogen procedures v1; vendors selected
- Jun 2027: H_2 handling install + E-stop architecture
- Sep 2027: First controlled ignition tests; repeatable starts
- Dec 2027: Evidence pack v1; certification roadmap active
- Apr 2028: Paid customer evaluation program launched
- Jul 2028: Series A closes; expand test capacity
- Dec 2028: Phase 1 maturity gate; 2029 plan locked

2029–2031 (annual gates)

- 2029: Certification-strength testing; manufacturing/QA system v1; Phase 2 concept freeze
- 2030: Aircraft program groundwork; integrated ground systems; ops playbook v2; ground test campaign
- 2031: Controlled flight demos (as authorized); early commercial agreements; support model operational

Team

Execution-focused leadership for deep-tech + safety-critical systems.

Jorge Peraza - Founder

Former Tesla Software Architect with experience building large-scale, safety-critical systems. Leads ColibriV's execution discipline across test evidence generation, tooling and infrastructure readiness, requirements traceability, and operational program delivery.

Advisors & partners (in development): propulsion, hydrogen safety, certification, and aerospace manufacturing experts.

Silvia Jimenez - Founder

BIM Manager and preconstruction specialist with **20+ years of experience** in construction project planning, budgeting, and operational execution across **civil works and architecture**. She brings deep expertise in turning complex projects into buildable, cost-controlled plans - supporting ColibriV's facility, infrastructure, and operational readiness in Guanacaste as the program scales.

Let's build the practical path to zero-carbon commercial flight.

Contact: invest@colibriv.com | colibriv.com